advocates, the long-run growth potential of stocks can be expected to materialize in time to keep the bond ladder adequately extended.

To understand this probability-based point of view, Exhibit 7.13 plots the historical worst-case annualized nominal returns for large-capitalization US stocks and intermediate-term US government bonds since 1926. Over shorter holding periods, bonds were less exposed to downside risks. Over one year, for instance, the worst case for bonds was a 5.1 percent drop, while stocks fell by 43.3 percent in their worst year. Over any historical three-year period, bonds provided a positive annualized return. It took fifteen years before stocks historically were always able to provide a positive return. But for holding periods of at least seventeen years, the historical worst-case annualized performance for stocks exceeds that for bonds. Over twenty-year periods, for instance, stocks experienced a worst-case 3.1 percent annualized return, compared to 1.6 percent for bonds. For thirty-year periods, the worst case for stocks was 8.5 percent, compared to 2.2 percent for bonds. And for forty-year periods, stocks' worst performance was an 8.9 percent annualized return, compared to 2.8 percent as the worst for bonds. For historical forty-year periods, even the best case for bonds (8.1 percent) could not beat the worst case for stocks. Because probability-based advocates have confidence that the historical record provides sufficient precedence for what can be expected in the future, this is the basic logic for understanding the view that stocks should be the primary asset designed to cover retirement expenses over the longer term.

Exhibit 7.13

Worst-Case Annualized Nominal Returns for Different Holding Periods

S&P 500 and Intermediate-Term Government Bonds, 1926–2016